

One leaked API key... and your entire production system could be compromised. So where do companies like Netflix, Uber, and Amazon safely store their secrets?



Why Configuration Management?



- Applications need configuration to run
- Examples: Database URL, API Keys, JWT Secrets
- Hardcoding secrets is insecure
- Centralized configuration simplifies management



What Are Secrets?



- Sensitive information used by applications
- API Keys, Passwords, Tokens, Certificates
- Must never be stored in source code
- Should be securely encrypted and controlled



What is HashiCorp Vault?



- Centralized secret management platform
- Securely stores and encrypts secrets
- Provides controlled access using authentication
- Supports dynamic and short-lived credentials



How Vault Works?



- Applications authenticate with Vault
- Vault verifies identity and permissions
- Authorized secrets are returned securely
- Secrets can be rotated automatically



Why Companies Use Vault?



- Improves security and compliance
- Eliminates hardcoded credentials
- Simplifies secret rotation
- Works across cloud and on-premises environments



Static vs Dynamic Secrets



- Static: Fixed passwords or API keys
- Dynamic: Temporary credentials generated on demand
- Dynamic secrets reduce security risks
- Automatically expire after use



Benefits of Centralized Configuration



- Single source of truth
- Easy environment management
- Better scalability
- Secure deployments
- Reduced human errors



Common Mistakes

